Expander Graphs in Cryptography

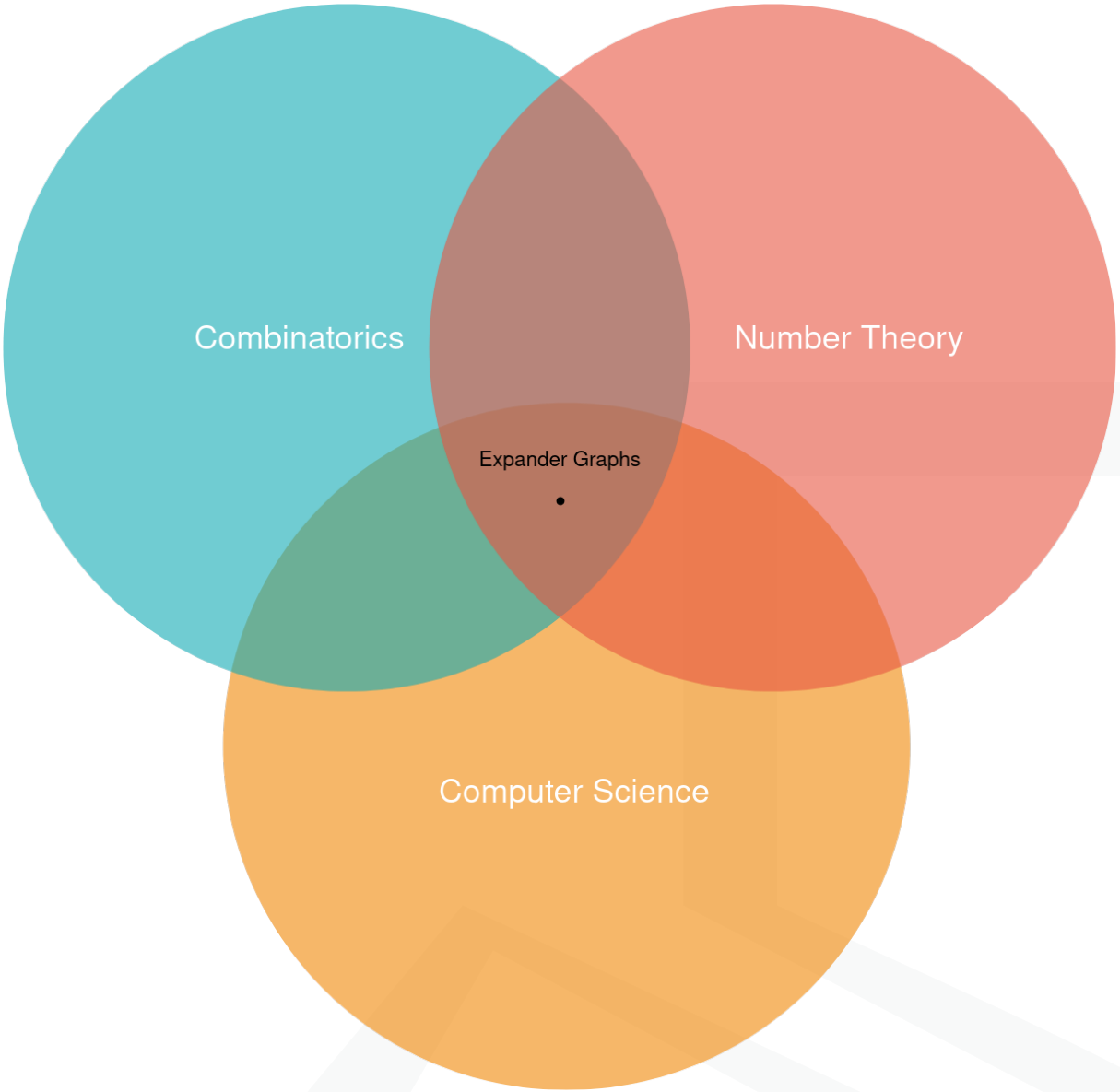
Shai Levin

Supervisor: Steven Galbraith

December, 2022

University of Auckland

Expander Graphs



Two Important Properties

On a 'good' expander graph:

- Path finding is hard when the number of vertices is (exponentially) large.
- Random walks converge to the uniform distribution.

Some Applications

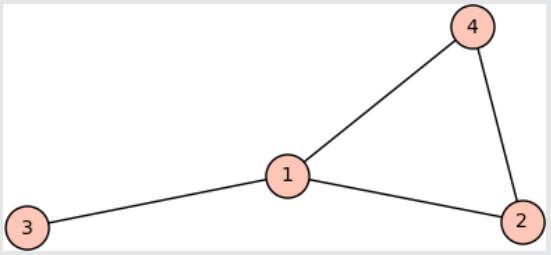
In Computer Science:

- Efficient error correcting codes
- Fault-tolerant networks
- Cryptographic pseudorandom behaviour:
  - Cryptographic hash functions or one way functions.
  - pseudorandom functions (?)

Adjacency Matrix, Regular graphs

An *adjacency matrix* A of a graph on n vertices is an $n \times n$ matrix where each $a_{i,j} = \#$ edges from i-th vertex to j-th vertex.

Example



For G above, $A = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$.

- A graph is d-regular if each vertex is adjacent to d others.
- We call a d-regular graph on n vertices an $[n, d]$-graph.

Graph Spectrum

- If a graph G is undirected, the adjacency matrix A is symmetric and real $\implies$ n eigenvalues.

Definition (Graph Spectrum)

The *spectrum* of a graph is the set of eigenvalues $\lambda_1, ...\lambda_n$ of adjacency matrix A where:

$$d \geq \lambda_1 \geq \lambda_2 \geq ... \geq \lambda_n \geq -d$$

for $d \in \mathbb{Z}$.

Graph Spectrum

- If a graph G is undirected, the adjacency matrix A is symmetric and real $\implies$ n eigenvalues.

Definition (Graph Spectrum)

The *spectrum* of a graph is the set of eigenvalues $\lambda_1, ...\lambda_n$ of adjacency matrix A where:

$$d \geq \lambda_1 \geq \lambda_2 \geq ... \geq \lambda_n \geq -d$$

for $d \in \mathbb{Z}$.

Some useful properties of graph spectrum:

- If G is d-regular, then $\lambda_1 = d$.
- G is bipartite if.f $\lambda_1 = -\lambda_n$.
- G is connected if.f $\lambda_1 > \lambda_2$.

Algebraic Definition: Spectral Gap

Given an $[n, d]$-graph G with spectrum $d = \lambda_1 \geq \lambda_2 \geq ... \geq \lambda_n \geq -d$:

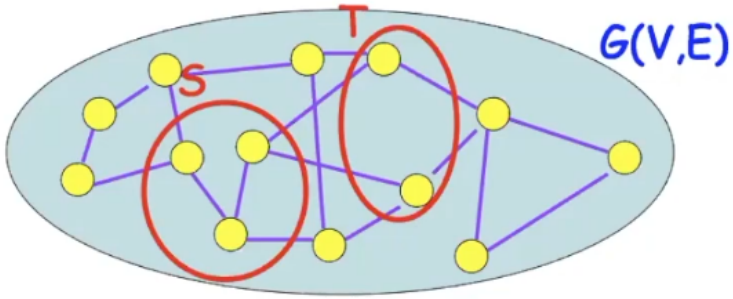
- Let $\lambda(G) := max(\{|\lambda_2|, ..., |\lambda_n|\})$.
- $d - \lambda(G)$ is the *spectral gap*.
- G is an $[n, d, \alpha]$-graph if $\lambda(G) \leq \alpha d$. Expander if $\alpha < 1$.
- An infinite family of graphs $\{G_i\}$ is an *expander family* if there exists an $\alpha \leq 1$ such that for all i, G_i is an $[n_i, d, \alpha]$ graph.

Expander Mixing Lemma

Theorem

The number of edges between any two large vertex subsets in a good expander graph, is close to the average amount of edges between two vertex subsets in a random $[n, d]$-graph.

- Corollary: random walks on the graph converge to uniform distribution in $O(log(n))$ steps.



Ramanujan Graphs

Definition (Ramanujan Graphs)

If G is an $[n, d]$-graph, then it is Ramanujan if:

$$2\sqrt{d-1} - \epsilon \leq \lambda(G) \leq 2\sqrt{d-1}.$$

for $\epsilon > 0$ where $\epsilon \to 0$ as $n \to \infty$.

- All connected graphs satisfy the lower bound.
- Smaller $\lambda(G)$ means a better expander.
- Ramanujan graphs are (asymptotically) the best expanders - but hard to come by.

Problem

Does there exist infinite families of d-regular Ramanujan graphs for each $d \geq 3$?

Constructing Expander Graphs

Two types of construction. Given an infinite family $\{G_i\}$ of $[n_i, d, \alpha]$-expander-graphs:

- **Weakly Explicit**: G_i can be constructed in polynomial time. (Polynomial in n_i).
- **Strongly Explicit**: Given $i \in \mathbb{N}$, a vertex $v \in V(G_i)$, the neighbours of v can be computed in polynomial time. (Polynomial in length of input (i, v)).

Expanders: What we know so far

- Ramanujan graphs are optimal expanders.
- Good expanders are sparse and finding paths on them is 'hard'.
- Can we use them in cryptography?

One Way Functions, Pseudorandom Generators and Pseudorandom Functions

Let $f : \mathcal{X} \to \mathcal{Y}$ be a function that is efficient to compute.

- f is *one way* if given f, some $f(x) \in \mathcal{Y}$, it is hard to compute an $x' \in \mathcal{X}$ such that $f(x') = f(x)$.
- f is a *pseudorandom generator* if the image of f is 'almost' uniform over $\mathcal{Y}$ (and $|\mathcal{X}| > |\mathcal{Y}|$)
- f is a *pseudorandom function* if f is indistinguishable from a uniform function when queried. (instantiated with a secret key)

One Way Functions, Pseudorandom Generators and Pseudorandom Functions

Let $f : \mathcal{X} \to \mathcal{Y}$ be a function that is efficient to compute.

- f is *one way* if given f, some $f(x) \in \mathcal{Y}$, it is hard to compute an $x' \in \mathcal{X}$ such that $f(x') = f(x)$.
- f is a *pseudorandom generator* if the image of f is 'almost' uniform over $\mathcal{Y}$ (and $|\mathcal{X}| > |\mathcal{Y}|$)
- f is a *pseudorandom function* if f is indistinguishable from a uniform function when queried. (instantiated with a secret key)

In fact, we don't even know if OWF/PRG/PRFs *exist*!

Applications: OWF, PRG, PRF

- **OWF**: Hash functions (missing collision resistance).
- **PRG**: Used to expand small amounts of randomness.
- **PRF** Act like keyed hash functions.
- **OWFs, PRFs:** Used for identification protocols or signature scheme. Using *zero knowledge proofs*: "Given pk, prove I know a secret sk such that $f(sk) = pk$."

One Way Function from Expanders

- On a 'good' $[n, d]$ expander graph G, $f : \{0, 1, .., d-2\}^k \to V(G)$ with an ordering on the vertex set:

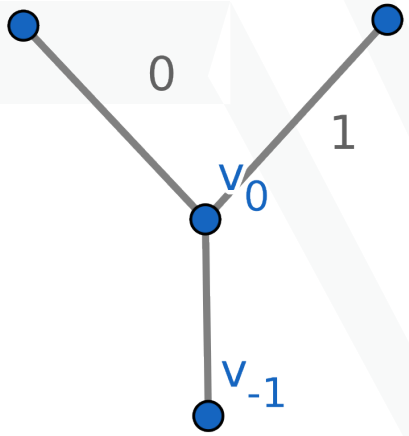
One Way Function from Expanders

- On a 'good' $[n, d]$ expander graph G, $f : \{0, 1, .., d-2\}^k \to V(G)$ with an ordering on the vertex set:
- Fix a starting v_0. Remember a previous vertex to prevent backtracking. For each step i of input, walk to the x_ith neighbour (that isn't v_{i-1}).

$f(\text{"01"})$ on a 3-regular graph

One Way Function from Expanders

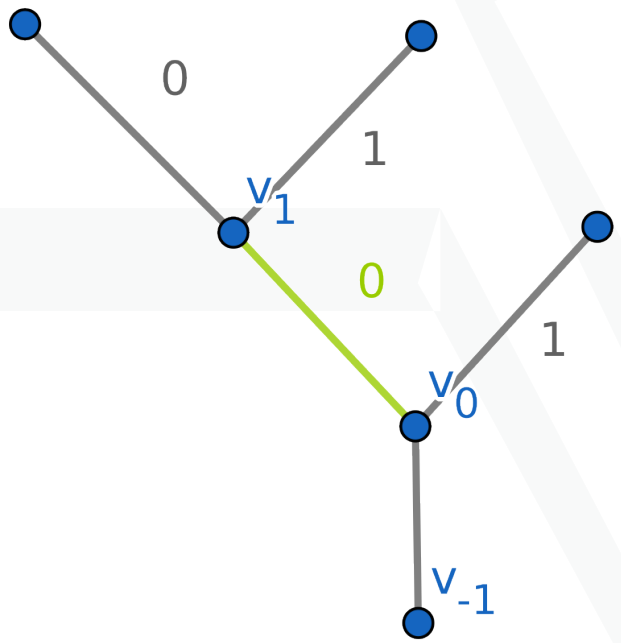
- On a 'good' $[n, d]$ expander graph G, $f : \{0, 1, .., d-2\}^k \to V(G)$ with an ordering on the vertex set:
- Fix a starting v_0. Remember a previous vertex to prevent backtracking. For each step i of input, walk to the x_ith neighbour (that isn't v_{i-1}).



f("01") on a 3-regular graph

One Way Function from Expanders

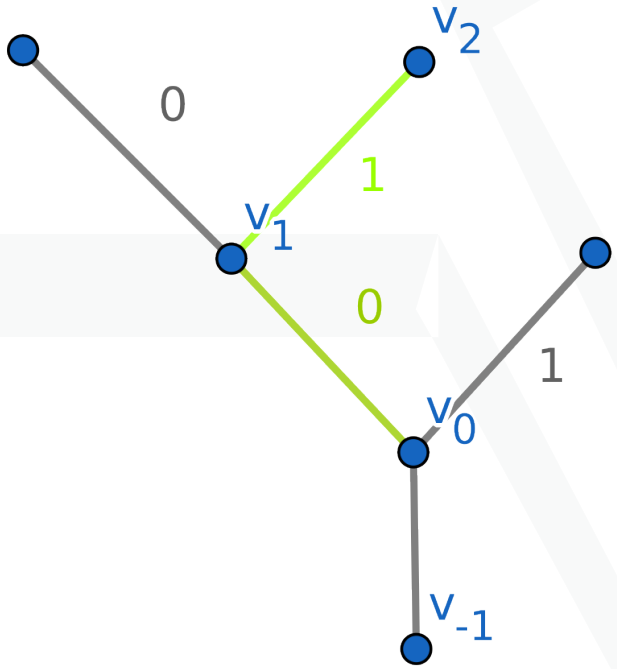
- On a 'good' $[n, d]$ expander graph G, $f : \{0, 1, .., d-2\}^k \to V(G)$ with an ordering on the vertex set:
- Fix a starting v_0. Remember a previous vertex to prevent backtracking. For each step i of input, walk to the x_ith neighbour (that isn't v_{i-1}).



$f(\text{"}01\text{"})$ on a 3-regular graph

One Way Function from Expanders

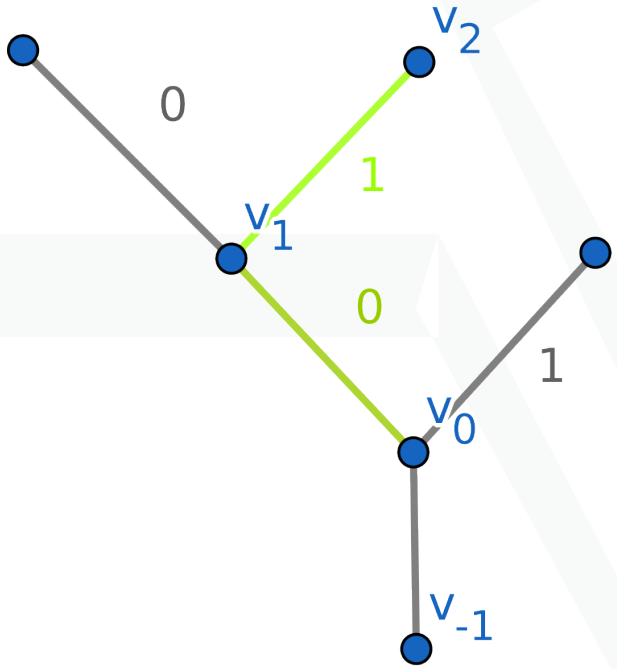
- On a 'good' $[n, d]$ expander graph G, $f : \{0, 1, .., d-2\}^k \to V(G)$ with an ordering on the vertex set:
- Fix a starting v_0. Remember a previous vertex to prevent backtracking. For each step i of input, walk to the x_ith neighbour (that isn't v_{i-1}).



$f(\text{"01"})$ on a 3-regular graph

One Way Function from Expanders

- On a 'good' $[n, d]$ expander graph G, $f : \{0, 1, .., d-2\}^k \to V(G)$ with an ordering on the vertex set:
- Fix a starting v_0. Remember a previous vertex to prevent backtracking. For each step i of input, walk to the x_ith neighbour (that isn't v_{i-1}).
- Ramanujan graph G is 'good' if:
  - strongly explicit,
  - exponentially large vertex set,
  - little symmetry,
  - hard to find cycles on.



f("01") on a 3-regular graph

Elliptic Curves and Isogenies

- **Elliptic curves**. $y^2 = x^3 + ax + b$ Algebraic and geometric structure. Set of solutions over a field form a group.
- Morphisms of elliptic curves are called **isogenies**. Isogenies preserve group & geometric structure.
- The degree of an isogeny is the size of it's kernel (as a group homomorphism).

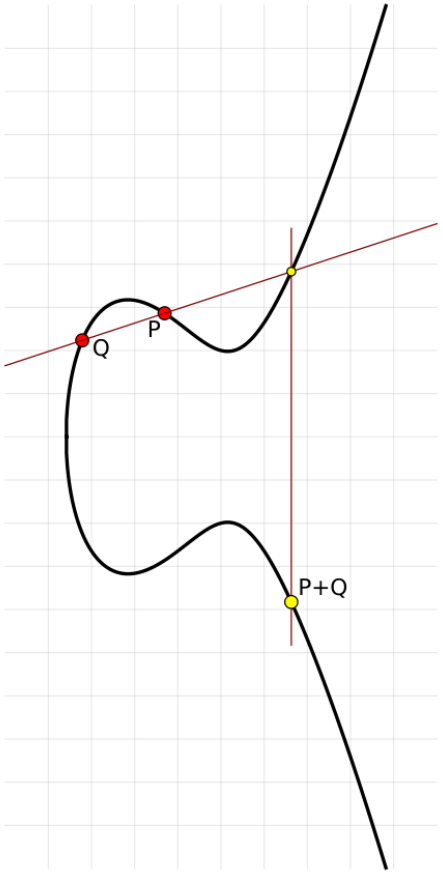


Figure 1: Group operation on $y^2 = x^3 - 4x + 7$

Supersingular ℓ-Isogeny graphs

- An elliptic curve is ordinary or supersingular.
- The set of supersingular elliptic curves defined over $\mathbb{F}_{p^2}$ (up to isomorphism) form a graph with degree ℓ isogenies as edges.
- The graph is 'good'. Ramanujan, $\lfloor \frac{p}{12} \rfloor$ vertices, $\ell + 1$ regular (for prime ℓ) and strongly explicit.
- Vertices are usually represented by the j-invariant, which corresponds to a single field element in $\mathbb{F}_{p^2}$ (1-1 correspondence)

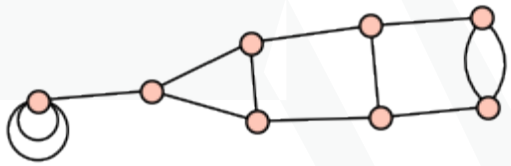


Figure 2: Supersingular isogeny graph $G_2(\mathbb{F}_{97^2})$

The CGL One Way Function

The CGL function below, where $G_2(\mathbb{F}_{p^2})$ is a 3-regular family of expanders for increasing p. For security, $p > 2^{256}$.

Algorithm 1 $h(m, j_0, j_{-1}, \mathrm{Sqr}(.))$

Input: An n bit binary string $m = m_{n-1}||...||m_0$, adjacent vertices j_0, j_{-1} in $G_2(\mathbb{F}_{p^2})$, and a deterministic square root algorithm Sqr for $\mathbb{F}_{p^2}$.

Output: Vertex j_n corresponding to the end point of the walk given by m.

```
for i in 0, ..., n − 1 do
    s_i ← +1 if m_i = 1, −1 if m_i = 0
    a_i ← −j_i^2 + 1488 j_i − 162000
    b_i ← 1488 j_i^2 + 40773375 j_i + 8748000000
    D_i ← (a_i + j_{i−1})^2 − 4(b_i + a_i j_{i−1} + j_{i−1}^2)
    S_i ← Sqr(D_i)
    j_{i+1} ← 2^{−1}(−a_i − j_{i−1} + s_i S_i)
end for
return j_n
```

Pseudorandom Function Candidate from CGL

Given a strongly explicit 3-regular expander family that is cryptographically secure, and a starting vertex v_0, $F_k(m)$ is computed as follows:

1. Walk the graph starting at v_0, based on the base-2 encoding of m, finishing on v_a
2. Walk the graph again, starting at v_a based on the base-2 encoding of k, finishing on v_b.

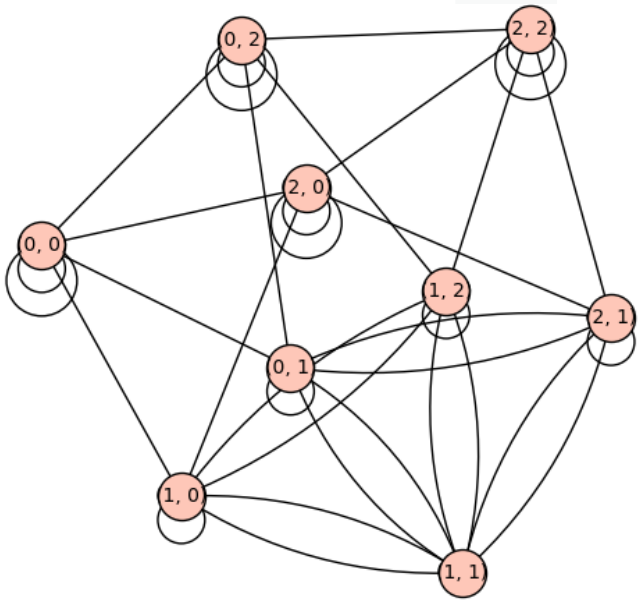
$$\begin{tikzcd}
v_{-1} \arrow[r, dash] & v_0 \arrow[r] \arrow[rrrr, bend left, dotted, "h(m,v_0,v_{-1})"] & v_1 \arrow[r] & \dots \arrow[r] & v_{a-1} \arrow[r] & v_a \arrow[r] \arrow[rrrr, bend left, dotted, "h(k,v_a,v_{a-1})"] & v_{a+1} \arrow[r] & \dots \arrow[r] & v_{b-1} \arrow[r] & v_b
\end{tikzcd}$$

Thank you!

Example: A Strongly Explicit Family of Expanders

Margulis Construction (Discrete Torus Expanders): A family of 8-regular graphs $G_i = (V_i, E_i)$ for $i \in \mathbb{Z}^+$:

- $V_i = \mathbb{Z}_i \times \mathbb{Z}_i$
- An edge from each vertex (x, y) to $(x \pm y, y)$, $(x, y \pm x)$, $(x \pm y + 1, y)$, $(x, y \pm x + 1)$. (arithmetic mod i)
- G_i is a $[i^2, 8, \frac{5\sqrt{2}}{8}]$ family of expanders where $\lambda(G_i) \to 2\sqrt{8-1}$ as $i \to \infty$.



PRF Reformulated: Vélu Formula Approach

We can reformulate the PRF in a similar way:

- Starting with an Elliptic Curve E over $\mathbb{F}_{p^2}$ where $p = 2^a \pm 1$
- Let P_0, Q_0 be a basis for the torsion subgroup $E[2^a]$.
- Define

$$F_k(m) : \mathcal{K} \times \mathcal{M} \to \mathbb{F}_{p^2}$$
$$(k, m) \mapsto E/\langle P_0 + [2^{\frac{a}{2}} k + m] Q_0 \rangle$$

- With message and key space $\mathcal{M} = \mathcal{K} = \mathbb{Z}_{2^{a/2}}$